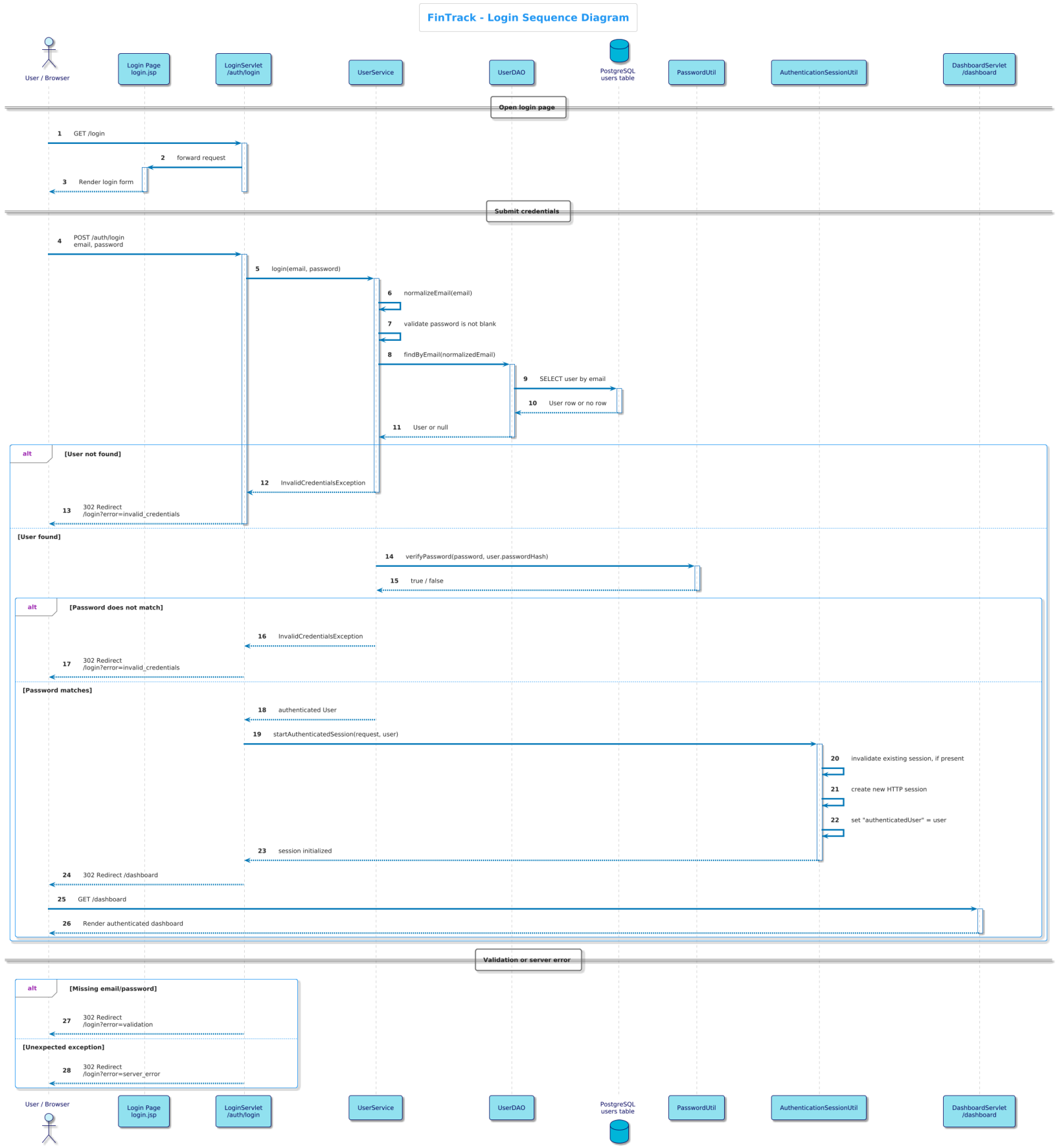

FinTrack - Login Sequence Diagram
User / Browser
Login Page login.jsp
LoginServlet /auth/login
UserService
UserDAO
PostgreSQL users table
PasswordUtil
AuthenticationSessionUtil
DashboardServlet /dashboard
Open login page
1 GET /login
2 forward request
3 Render login form
Submit credentials
4 POST /auth/login email, password
5 login(email, password)
6 normalizeEmail(email)
7 validate password is not blank
8 findByEmail(normalizedEmail)
9 SELECT user by email
10 User row or no row
11 User or null
alt
[User not found]
12 InvalidCredentialsException
13 302 Redirect /login?error=invalid_credentials
[User found]
14 verifyPassword(password, user.passwordHash)
15 true / false
alt
[Password does not match]
16 InvalidCredentialsException
17 302 Redirect /login?error=invalid_credentials
[Password matches]
18 authenticated User
19 startAuthenticatedSession(request, user)
20 invalidate existing session, if present
21 create new HTTP session
22 set "authenticatedUser" = user
23 session initialized
24 302 Redirect /dashboard
25 GET /dashboard
26 Render authenticated dashboard
Validation or server error
alt
[Missing email/password]
27 302 Redirect /login?error=validation
[Unexpected exception]
28 302 Redirect /login?error=server_error